Test Assignment for Engineering Lead (DeFi & Crypto)

Task
Build a minimal working application where a user:
• Deposits SOL as collateral on Kamino
• Borrows USDC
• Bridges the USDC to Base via Mayan to be held in a separate Base wallet
• Closes the position by bridging USDC back to Solana and repaying in full

Fixed Parameters (so you can focus on integrations rather than configuration):
• Kamino main market
• $20 SOL collateral (please email us to send funds to your address of choice if needed)
• $5 USDC borrow amount
• Solana + Base only
• Mayan for bridging. Slippage and fees are not important for this assignment

What to Deliver
• Git repository
• AI tooling writeup
• Time log
• Short video walkthrough on call with KAST team covering what you built
• Discussion with KAST team on how you would scope, expand, and implement this in production

Key Priorities
• **The main goal is a working happy path end-to-end.** Everything else (UI/UX, error handling, security, tests, edge cases) is secondary.
• **This is designed to be an unreasonably large task**. You are expected to actively use AI agents. We specifically want to see how you orchestrate multiple agents in parallel with clear task boundaries. Candidates with a combination of domain knowledge and strong AI workflows will perform well in this task.
• Wallet and signing approach is your choice (this is not being tested).

Optional Bonuses (do these only if you have time)
• Use Privy for wallets
• Enable user defined partial repayments versus full repayment
• Enable collateral adjustments to add or withdraw collateral from Kamino
We value depth in one or two bonuses rather than breadth across all of them.

Time
Target is 3-6 hours. Please send what you have at the time box if you don’t finish everything. This is completely acceptable and expected.